

AYUSH KUMAR

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EDUCATION

Bachelor of Computer Science, SRM IST University

2023 - 2027

SKILLS

Languages	Python, Java, C, C++
Frameworks/Libraries	Scikit-Learn, Pandas, NumPy, Matplotlib, Flask, PyTorch, RAG
Tools	Android Studio, Unity3D, VS-Code, Git
Soft Skills	Leadership , Project Head, Decision Making, Collaboration

EXPERIENCE

Research Intern AI-Driven Mineral Deposit Prediction	Dec 2024 – Present <i>IIT Bombay</i>
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- Curating a comprehensive dataset of existing mineral deposits.
- Developing machine learning models to predict potential mineral deposit locations based on various geological and geospatial factors.

Research Intern AI-Driven Multimodal Framework for Predicting and Managing Ocean Deoxygenation	Oct 2024 - Present <i>SRM IST</i>
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- **Best Paper Award at the International Conference on AI for the Oceans 2025**
- Developed a machine learning pipeline to predict dissolved oxygen levels and classify low oxygen (hypoxic) areas in marine environments.
 - Leveraged advanced deep learning techniques, including CNN and convLSTM, for regression and classification tasks.
 - Aimed to assist ecological research and decision-making by identifying dead zones where aquatic life cannot thrive due to hypoxia, achieving 97% accuracy

Technical Lead NextGenAI	Sept 2023 - Present <i>SRM IST</i>
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- Helped organize Pentathlon 2024, coordinating over 200 participants in an 5-stream hackathon.
- Organized a Bootcamp for 30+ club members, resulting in a 25% increase in ML expertise among attendees.
- Contributed to a Machine Learning research project for the Biotech department, leveraging Python and Scikit-learn to clean and create an ensemble model for a Large Dataset.

Technical Associate MLSA	Sept 2023 - Present <i>SRM IST</i>
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- Gained hands-on experience with Large Language Models (LLMs) and LangChain, implementing advanced NLP techniques for language understanding.

PROJECTS

PDF Data Extractor

- Developed an Android application for PDF data extraction using EasyOCR, Tesseract, and Flask, enabling users to upload documents and retrieve text efficiently.

Dino Game using Gesture

- Developed a gesture-based control system for Chrome's Dino Game using OpenCV and Python, allowing players to control the game with hand movements, enhancing accessibility.

Hack RX

- Contributed to the development of an Agentic LLM by designing a Retrieval-Augmented Generation (RAG) architecture for storing and processing text, images, and documents, improving system flexibility and performance.

Bitcoin Price Predictor

- Built an ML model using historical Bitcoin data (opening price, high/low price) to predict closing prices, leveraging Python, Pandas, and Scikit-Learn for data preprocessing and modeling.

LEADERSHIP

- Technical Lead at [NextGenAI](#)
As Technical Lead at NextGenAI, conducted Boot Camp for our Club members to garner their interest in Machine Learning and simultaneously expanding their repertoire.
- Technical Associate at [MLSA](#)
- Technical Member at [NSCC](#)

CERTIFICATIONS

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| • Supervised Machine Learning: Regression and Classification | Stanford University, Coursera |
| • Unsupervised Learning, Recommenders, Reinforcement Learning | Stanford University, Coursera |
| • Advanced Learning Algorithms | Stanford University, Coursera |
| • Neural Networks and Deep Learning | DeepLearning.AI |

EXTRA-CURRICULAR ACTIVITIES

- Actively participated in weekly reading sessions at SRM University, engaging in discussions on novels and contemporary literature.